Lab 07 – Reflection – Alexander DeBusk

During this lab, I learned about different ways to evaluate a machine learning model and understand how well it makes predictions. One of the main things I learned was how to use a confusion matrix. It helped me see the difference between correct and incorrect predictions, including false positives and false negatives. At first, it was a little confusing to understand what each part meant, but looking at the heatmap made the results easier to understand.

Another important concept was precision, recall, and the F1-score. I learned that each metric measures a different part of the model's performance. For example, precision is important when we want to avoid false alarms, while recall is important when missing a real case could be dangerous. These metrics can be usefull in real world situations, such as detecting spam emails or identifying possible medical problems. I also learned that accuracy alone does not always tell us everything about a model.

Cross-validation was another technique that helped me understand model evaluation better. Instead of relying on just one train-test split, the model is tested using multiple folds of the dataset. This can provide a more reliable estimate of how the model might perform on new data. One challange I encountered was organizing the code because some of the notebook code was all placed on one line. I had to seperate the statements properly so that the code could be read and executed correctly.

These techniques are important in generative AI because models also need to be evaluated to determine how reliable their outputs are. Developers can use evaluation methods to

identify weaknesses, compare different models, and make improvements. Learning these concepts gave me a better understanding of why testing a model involves more than just checking if its predictions are correct.

REFLECTION JOURNAL – **Module 07: Model Evaluation**

ITAI 1371

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In this module, I learned how to evaluate a machine learning model using the Titanic dataset and Logistic Regression.

The confusion matrix helped me understand correct and incorrect predictions, including true positives, true negatives, false positives, and false negatives. I also learned how precision, recall, and F1-score provide more information about model performance than accuracy alone.

Through five-fold cross-validation, I obtained an average score of 78.68%, with a standard deviation of 0.0094, helping me understand how consistently the model performed across different data splits.

One challenge I faced was a Python NameError caused by a missing Pandas import. I corrected the error and successfully ran the code. Overall, this module improved my understanding of model evaluation, troubleshooting, and interpreting machine learning results.

Reflection

In this lab, I learned how to use confusion matrices, precision, recall, F1 scores, and cross validation with the Titanic dataset. This lab helped me better understand how to check a machine learning model and gave me some useful ideas. Before this lab, I mainly focused on accuracy when checking a model. I learned that a high accuracy score does not always mean the model works well. For example, a medical model might have high accuracy if it predicts that everyone is healthy, but it could miss many people who have a serious disease. This showed me that I need to look at different scores based on what the model is used for. The confusion matrix and classification report helped me understand what kinds of mistakes a model makes.

I learned that precision and recall are both important, but their importance depends on the situation. For example, high precision is important in fraud detection because wrongly marking a normal transaction as fraud can cause problems for customers. In medical tests, high recall is important because missing a person with a serious disease can be dangerous. This helped me understand why we need to choose evaluation methods based on the problem. Before this lab, I mainly used a single train test split to evaluate models. I learned that the results could change depending on how the data is divided. If the test data is too easy or too difficult, the score may not show how well the model really works. Cross validation tests the model on different parts of the dataset and calculates the average score. I now understand that this method can give a more reliable result and help me better evaluate machine learning models.